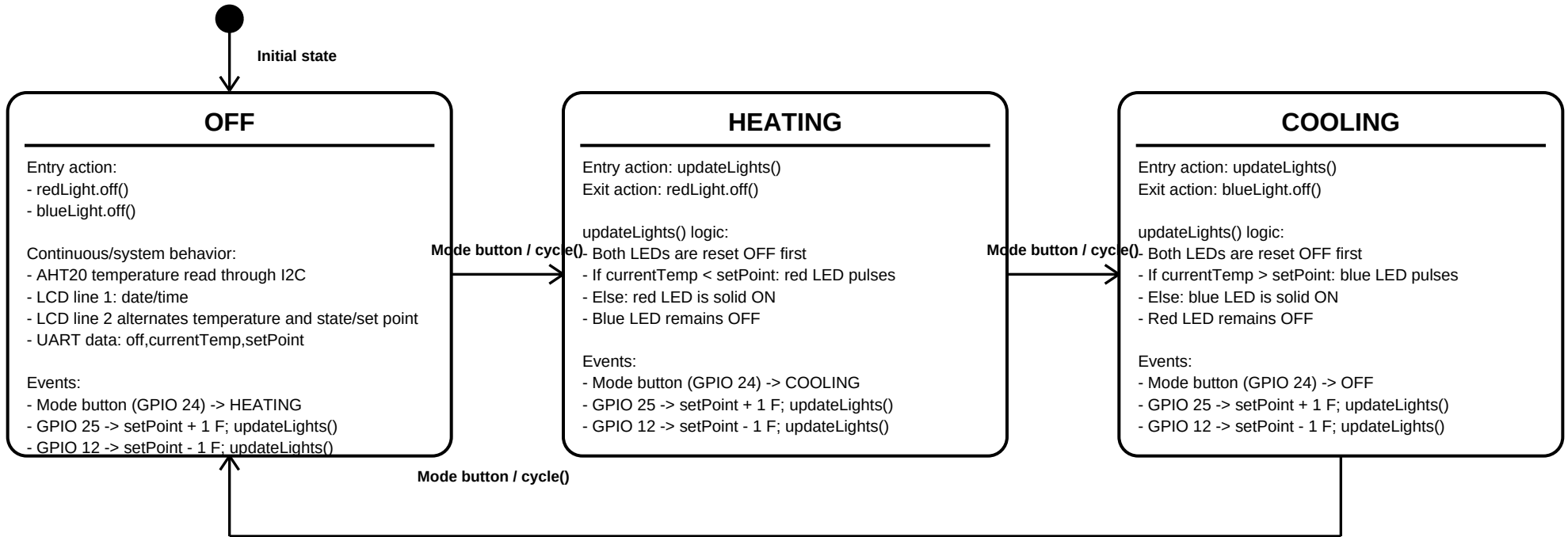


CS 350 Final Project - Smart Thermostat State Machine

Implementation-aligned documentation for Thermostat.py



System-Wide Technical Specification

- Initial set point = 72 F; state machine initializes in OFF and cycles OFF -> HEATING -> COOLING -> OFF.
- AHT20 sensor uses I2C; getFahrenheit() converts Celsius readings to Fahrenheit.
- GPIOZero buttons: GPIO 24 changes mode, GPIO 25 increases set point, GPIO 12 decreases set point.
- LCD line 1 displays date/time; line 2 alternates current Fahrenheit temperature with state and set point.
- UART is initialized at 115200 baud, 8 data bits, no parity, 1 stop bit, with a 1-second timeout.
- Every 30 seconds, UART sends a comma-delimited record: state,current temperature,set point.
- LED status is recalculated on state entry and after either set-point button changes the target temperature.